Root distribution in the Kadar–Yu Gram sequences

Research proof and Lean formalization
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Abstract

We prove squarefreeness, eventual stabilization of the exterior root count, and density in $[-2,2]$ for the reduced Gram-polynomial sequences appearing in Morris–Martin Conjecture 4.5.1. The argument applies to every finite real symmetric head with one appended free path. Removing the monic gcd of the first two determinants gives the relevant reduced sequence. A codimension-one eigenvalue-multiplicity bound proves squarefreeness, including the boundary index; spectral counts give stabilization and density. The universal Gram model and its complex coefficient and root-multiset bridges are verified in Lean. The source application uses the mixed Gram structure and factorization already established by Morris and Martin.

1 The Gram model and the source conclusion

Let G be a real symmetric $m\times m$ matrix, $m\ge 1$, with a distinguished coordinate placed last. Form G_s by adjoining a path of s new vertices: the new diagonal is zero, each adjacent path coupling is 1, and the first new vertex is coupled only to the distinguished coordinate, also by 1. Every entry of the original head G is retained. Put

$$f_s(X)=\det(XI+G_s),\qquad g=\operatorname*{gcd}_{\text{monic}}(f_0,f_1),\qquad p_s=f_s/g\quad(s\ge 0),\qquad p_{-1}=Xp_0-p_1. \tag{1}$$

All gcds and quotients can be taken over either $\mathbb{R}$ or $\mathbb{C}$; the resulting monic polynomials agree under coefficient extension. Roots are counted with multiplicity throughout.

Theorem 1. *Every p_s, $s\ge -1$, is monic, real-rooted, and squarefree. There are integers $S,C\ge 0$, with $C\le 2m$, such that p_s has exactly C roots outside $[-2,2]$ whenever $s\ge S$. Moreover,*

$$[-2,2]\subseteq\overline{\bigcup_{s\ge 0}\{x\in[-2,2]:p_s(x)=0\}}.$$

The head G may have arbitrary internal couplings.

Corollary 2. *Fix $l\ge -1$ and a partition $\lambda\vdash l+2$. For the reduced sequence $P_n^{(\lambda)}$ of Morris–Martin [1, Theorem 3.19], every $P_{l+4+k}^{(\lambda)}$, $k\ge -1$, is squarefree. Its number of roots outside $[-2,2]$ is eventually constant as $k\to\infty$, and*

$$\bigcup_{k\ge 0}\{\alpha\in[-2,2]:P_{l+4+k}^{(\lambda)}(\alpha)=0\}\quad\text{is dense in }[-2,2].$$

Thus all three assertions of [1, Conjecture 4.5.1] hold.

The source identification proving the corollary is given in Section 5. The classical spectral and polynomial tools below receive no separate novelty claim.

2 Reduction and the boundary polynomial

Write $H_s=-G_s$, so f_s is the genuine characteristic polynomial of H_s. The appended path couplings in H_s are -1. Expansion at a terminal vertex gives

$$f_{s+2}=Xf_{s+1}-f_s. \tag{2}$$

Consequently g divides every f_s. The quotients p_s are monic, and

$$p_{s+2}=Xp_{s+1}-p_s,\qquad \gcd(p_s,p_{s+1})=1. \tag{3}$$

The initial coprimality follows by removing the actual gcd; the recurrence propagates it. Since each p_s divides a real symmetric characteristic polynomial, each splits over $\mathbb{R}$.

Let H_- be H_0 with the distinguished coordinate deleted. The first border identity is

$$f_1=Xf_0-\det(XI-H_-),\qquad \det(XI-H_-)=g(Xp_0-p_1)=gp_{-1}. \tag{4}$$

This proves that p_{-1} is a monic, real-split polynomial as well. It is coprime to p_0 by (3). The 0×0 minor has characteristic polynomial 1, so the statement includes $m=1$. No coprimality between g and a quotient is used.

3 Spectral counts and squarefreeness

For a real symmetric matrix A, let $N_A^+(t)$ and $N_A^-(t)$ count its eigenvalues strictly above and strictly below t, respectively.

Lemma 3. *If B is a principal submatrix of A obtained by deleting r coordinates, then, for either sign,*

$$N_B^{\pm}(t) \le N_A^{\pm}(t) \le N_B^{\pm}(t) + r.$$

For $r = 1$, the multiplicities of any real t as a root of the two characteristic polynomials differ by at most one.

Proof. For the upper bound, project the span of the eigenvectors of A with eigenvalues greater than t onto the deleted coordinates together with the greater-than-t eigenspace of B. This projection is injective: a nonzero vector in its kernel would have quadratic form strictly greater than t times its squared norm when evaluated with A, and at most that quantity when evaluated with B. Thus its dimension is at most $r + N_B^+(t)$. The corresponding positive subspace of B, extended by zeros, gives the lower bound. Apply the same argument to $-A, -B$ for the other sign.

For $r = 1$, put $\delta_\pm = N_A^{\pm}(t) - N_B^{\pm}(t) \in \{0, 1\}$. If $m_A(t), m_B(t)$ denote the root multiplicities, counting the entire spectrum yields

$$m_A(t) - m_B(t) = 1 - \delta_+ - \delta_- \in \{-1, 0, 1\}.$$

□

Apply the multiplicity bound to H_s and H_{s+1}. If t is a root of p_s, it is not a root of p_{s+1} by (3). Hence

$$\mathrm{mult}_t(p_s) = \mathrm{mult}_t(f_s) - \mathrm{mult}_t(f_{s+1}) \le 1,$$

because the fixed factor g contributes the same multiplicity to both. Real splitting proves that p_s is squarefree. Applying the same argument to the principal pair H_-, H_0, using (4) and $\gcd(p_{-1}, p_0) = 1$, proves squarefreeness at $s = -1$. This proves the first assertion of Theorem 1.

4 Exterior stabilization and interior density

The free path on s vertices has the distinct eigenvalues

$$2\cos\frac{j\pi}{s+1}, \qquad 1 \le j \le s. \tag{5}$$

Changing every path coupling from 1 to -1 gives the same characteristic polynomial. All eigenvalues in (5) lie strictly between -2 and 2. The path is a principal submatrix of H_s with m deleted head coordinates. Lemma 3 therefore bounds both $N_{H_s}^+(2)$ and $N_{H_s}^-(-2)$ by m. Each is nondecreasing when a new vertex is appended. They are bounded integer sequences, so both eventually stabilize. The multiset identity

$$\mathrm{Roots}(f_s) = \mathrm{Roots}(g) + \mathrm{Roots}(p_s) \tag{6}$$

then subtracts a fixed exterior count and proves the stabilization claim. The strict thresholds exclude the endpoints, as required for roots outside the closed interval.

For density, fix $-2 < a < b < 2$ and choose $a < B < b$. The exact grid (5) supplies arbitrarily many distinct path eigenvalues in (a, B). For completeness, if $0 < u < v < 1$, choose M so large that $(R+1)/(M+1) < v - u$. The R consecutive indices starting at $\lfloor u(M+1) \rfloor + 1$ give grid fractions in (u, v). Apply this with $u = \arccos(B/2)/\pi$ and $v = \arccos(a/2)/\pi$.

Choose a path having at least $m + \deg g + 1$ such eigenvalues. Let T_s be its matrix. For a symmetric matrix A, let $I_A(a, B) = N_A^+(a) - N_A^+(B)$, the number of eigenvalues in $(a, B]$. Lemma 3 at both thresholds gives

$$I_{H_s}(a, B) \ge I_{T_s}(a, B) - m \ge \deg g + 1. \tag{7}$$

At most $\deg g$ of these roots can be contributed by g, counting multiplicity. By (6), p_s has a root in $(a, B] \subset (a, b)$. Every open subinterval of $(-2, 2)$ is therefore hit. Passing to closure and using $\overline{(-2, 2)} = [-2, 2]$ includes both endpoints. This completes Theorem 1.

5 Identification of the source sequence

Set $N = l + 4$ and $d = \dim S^\lambda \geq 1$. Use the real orthonormal Specht basis in [1, Section 3.2.3]. In the mixed family of [1, (3.17)], fix $d-1$ tail parameters at N and let the remaining parameter be $N+s$. Reorder the attaching vector $u_{l+3,l+4}b_d$ last in the initial matrix. Equations (3.2.3), (3.2.4), and the literal border (3.2.17) show that the resulting Gram matrices are exactly $XI + G_s$ for one fixed finite real symmetric head G. All interactions between different heads remain in G.

Proposition 3.25 of [1] gives, in this orthonormal convention,

$$f_s = C^{(\lambda)}\big(P_N^{(\lambda)}\big)^{d-1}P_{N+s}^{(\lambda)} = FP_{N+s}^{(\lambda)}, \tag{8}$$

where F is fixed and monic. Reducedness and the minus recurrence imply $\gcd(P_N^{(\lambda)}, P_{N+1}^{(\lambda)}) = 1$. Thus the monic gcd of f_0, f_1 is exactly F, including all of its multiplicities, and

$$p_s = P_{N+s}^{(\lambda)} \quad (s \geq 0), \qquad p_{-1} = P_{N-1}^{(\lambda)}.$$

The construction agrees with the maximal-common-factor reduction in [1, (3.2.12)]. The source's actual determinant recurrences (3.2.7) and (3.2.9) have the minus sign used throughout this proof. Theorem 1 proves Corollary 2 for every partition and all $k \geq -1$.

6 Formal certificate and scope

The accompanying Lean development contains 190 declarations in 22 modules. Its closed endpoints are

`gramModelReduced_root_distribution`
`gramModelReducedComplex_root_distribution`.

They quantify over every finite real symmetric head and use the literal determinant and normalized gcd quotients in (1). No factorization, consecutive coprimality, cyclic-compression, or spectral-bound assertion is left as a premise of these endpoints.

The complex development defines the quotient using the gcd over $\mathbb{C}$, proves its equality to the real quotient under coefficient mapping, transfers the entire root multiset, proves every complex root real, and verifies the exterior count and density with actual complex evaluation. The boundary polynomial is included. A separate monic-factor uniqueness lemma identifies any complex source factorization (8) with this quotient sequence.

The source application in Section 5 invokes the mixed-Gram structure and Proposition 3.25 already proved in [1]; Specht modules are not reimplemented in the certificate. The formal theorem proves the universal matrix model that contains those mixed families. Source fidelity and the scope of this passage are recorded explicitly in the accompanying review materials. The strict replay uses Lean 4.34.1 and mathlib revision `d13f23b7`, compiles every module afresh with warnings as errors, and checks every declaration's axiom dependencies. Only `propext`, `Classical.choice`, and `Quot.sound` occur.

The proposed source resolution is one combined outcome. Cauchy comparison, Chebyshev formulas, gcd reduction, the complex scalar extension, and individual clauses receive no separate credit. The source interpretation and a bounded prior-art investigation remain subject to independent review; this draft makes no claim of absolute worldwide priority.

References

[1] B. Morris and P. Martin, *On semisimplicity criteria and non-semisimple representation theory for the Kadar-Yu algebras*, arXiv:2512.24535v1 (31 December 2025), https://arxiv.org/abs/2512.24535.